

# Security+ 4.5 · Secure Protocols

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 4.5 — Secure Protocols (4:47)** · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

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**Executive Lesson Summary:** Secure protocols use encryption and authentication to protect data in transit. Secure: SSH (22), HTTPS (443), SFTP (22), SNMPv3, LDAPS (636), TLS 1.2/1.3, IPsec, WPA3. Insecure: Telnet (23), FTP (21), HTTP (80), SNMPv1/v2, LDAP (389), WEP/WPA, SMTP (25 without TLS).

## High-Yield Core Concepts Tested on Exam

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- Secure ports: SSH 22, HTTPS 443, SFTP 22, SNMPv3, LDAPS 636
- Insecure protocols (prohibited in exam): Telnet 23, FTP 21, HTTP 80, SNMPv1/v2
- TLS 1.2/1.3, IPsec, WPA3 are secure alternatives
- LDAP 389 is insecure; LDAPS 636 is secure
- SMTP 25 without TLS is insecure

## Exam Tips & Classic Scenarios

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- 'replaces telnet' = SSH (remote secure)
- If asked for secure remote protocol on port 22, answer SSH, not Telnet

**Typical Exam Scenario:** A question asks for a secure remote protocol on port 22. The correct answer is SSH, not Telnet.

## Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 4.5 — Secure Protocols)

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**Why is this advanced coverage required for SY0-701?** Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **TACACS+:** Terminal Access Controller Access-Control System Plus: Enterprise AAA protocol running over TCP port 49; fully encrypts the entire packet payload and decouples authentication from authorization, enabling granular per-command control on network switches and routers.
- **RADIUS:** Remote Authentication Dial-In User Service: Client/server AAA protocol (UDP ports 1812/1813) standard in 802.1X and VPN environments; encrypts only the password field and combines authentication and authorization.
- **KERBEROS:** Default Active Directory authentication protocol based on symmetric ticket granting issued by the Key Distribution Center (KDC: AS and TGS), preventing cleartext credentials on the wire.
- **LDAP:** Lightweight Directory Access Protocol: Protocol for querying centralized directory services (cleartext over TCP port 389).
- **LDAPS:** LDAP over SSL/TLS: Encrypted directory service queries using X.509 digital certificates over TCP port 636.

### **Classic CompTIA Traps & Question Framing:**

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

## **Practice Questions with In-Depth Solutions (10 Questions)**

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Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

**1. Secure protocol to administer a server via remote command line, on port 22:**

- A) Telnet**
- B) SSH**
- C) FTP**
- D) HTTP**

**2. Which authentication protocol uses tickets and is widely used in Active Directory environments?**

- A) RADIUS**
- B) TACACS+**
- C) Kerberos**
- D) LDAP**

**3. Corporate scenario (Secure Protocols):** During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'Secure ports: SSH 22, HTTPS 443, SFTP, SNMPv3, LDAPS 636'. What is the best recommendation or applicable control classification?

- A) Temporarily disable audit logs to accelerate system throughput**
- B) Replace preventive technical controls with informal terms of adherence**
- C) Implement the corresponding solution: Secure ports: SSH 22, HTTPS 443, SFTP, SNMPv3, LDAPS 636.**
- D) Allow shared access among administrators using generic credentials**

**4. Corporate scenario (Secure Protocols):** During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'INSECURE (prohibited on the exam): Telnet 23, FTP 21, HTTP 80, SNMPv1/v2.' What is the best recommendation or applicable control classification?

- A) Temporarily disable audit logs to accelerate system throughput**
- B) Implement the corresponding solution: INSECURE (prohibited on the exam): Telnet 23, FTP 21, HTTP 80, SNMPv1/v2.**
- C) Replace preventive technical controls with informal terms of adherence**
- D) Allow shared access among administrators using generic credentials**

**5. Corporate scenario (Secure Protocols):** During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'replaces telnet'=SSH (remote secure)'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: 'replaces telnet'=SSH (remote secure)...

**6. Corporate scenario (Secure Protocols):** During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'secure remote protocol on port 22' = SSH; not Telnet.' What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'secure remote protocol on port 22' = SSH; not Telnet...
- D)** Allow shared access among administrators using generic credentials

**7. Corporate scenario (Secure Protocols):** During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'Secure ports: SSH 22, HTTPS 443, SFTP, SNMPv3, LDAPS 636'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Secure ports: SSH 22, HTTPS 443, SFTP, SNMPv3, LDAPS 636...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

**8. Corporate scenario (Secure Protocols):** During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'INSECURE (prohibited on the exam): Telnet 23, FTP 21, HTTP 80, SNMPv1/v2.' What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: INSECURE (prohibited on the exam): Telnet 23, FTP 21, HTTP 80, SNMPv1/v2...

**9. Corporate scenario (Secure Protocols):** During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'replaces telnet'=SSH (remote secure)'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'replaces telnet'=SSH (remote secure)...
- D)** Allow shared access among administrators using generic credentials

**10. Corporate scenario (Secure Protocols):** During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'secure remote protocol on port 22'=SSH; not telnet.'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'secure remote protocol on port 22'=SSH; not telnet....
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

## COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

### 1) B — SSH.

✓ **Why it is correct:** SSH (port 22) encrypts the remote session. Telnet (23), FTP (21), and HTTP (80) send data in cleartext — insecure.

#### ✗ **Why the others are wrong:**

- *Telnet*: Legacy unencrypted protocol that transmits credentials and data in cleartext, vulnerable to network sniffing.
- *FTP*: Legacy unencrypted protocol that transmits credentials and data in cleartext, vulnerable to network sniffing.
- *HTTP*: Legacy unencrypted protocol that transmits credentials and data in cleartext, vulnerable to network sniffing.

### 2) C — Kerberos.

✓ **Why it's correct:** **Kerberos** is a ticket-based authentication protocol widely used in Microsoft Active Directory (AD) environments. It uses symmetric cryptography and a Key Distribution Center (KDC) to authenticate users and services without transmitting passwords over the network. The process involves obtaining a Ticket Granting Ticket (TGT) and subsequently service tickets to access resources. In the CompTIA Security+ (SY0-701) syllabus, Kerberos is highlighted as the default authentication protocol in Windows domains, offering mutual authentication and single sign-on (SSO) support. The mention of 'tickets' and 'Active Directory' points directly to Kerberos.

#### ✗ **Why the others are wrong:**

- *RADIUS*: RADIUS (Remote Authentication Dial-In User Service) is an authentication, authorization, and accounting (AAA) protocol used primarily for remote network access, such as VPNs, corporate Wi-Fi, and dial-up connections. It does not use tickets; instead, it employs a client-server model with attributes and challenges (such as PAP, CHAP, EAP). Although it can integrate with Active Directory to validate credentials, it is not the native AD authentication protocol and is not ticket-based. Therefore, it does not match the description.
- *TACACS+*: TACACS+ (Terminal Access Controller Access-Control System Plus) is an AAA protocol developed by Cisco, used for authentication, authorization, and accounting of network devices (routers, switches). It separates AAA functions and encrypts the entire payload, but does not use tickets and is not the default authentication mechanism for Active Directory. Its focus is managing access to infrastructure equipment, not authenticating users in Windows domains. Thus, it is incorrect for the presented context.
- *LDAP*: LDAP (Lightweight Directory Access Protocol) is a protocol for accessing directory services, such as Active Directory, allowing queries and modifications to objects (users, groups, computers). It is used for reading and writing to the directory, but is not a ticket-based authentication protocol. Authentication in AD typically occurs via Kerberos or NTLM; LDAP can be used for simple authentication (bind), but does not employ tickets and is not the primary authentication mechanism in AD environments. Therefore, it does not meet the 'uses tickets' description.

### 3) C — Implement the corresponding solution: Secure ports: SSH 22, HTTPS 443, SFTP, SNMPv3, LDAPS 636..

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Secure Protocols): Secure ports: SSH 22, HTTPS 443, SFTP, SNMPv3, LDAPS 636. These protocols provide encryption and integrity for remote administration, web traffic, file transfer, network management, and directory services, replacing insecure counterparts such as Telnet, HTTP,

FTP, SNMPv1/v2, and LDAP.

**✗ Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

**4) B — Implement the corresponding solution: INSECURE (prohibited on the exam): Telnet 23, FTP 21, HTTP 80, SNMPv1/v2..**

**✓ Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Secure Protocols): INSECURE (prohibited on the exam): Telnet 23, FTP 21, HTTP 80, SNMPv1/v2. These protocols transmit data in cleartext and are considered insecure. The best recommendation is to replace them with secure alternatives such as SSH (22), SFTP/FTPS, HTTPS (443), and SNMPv3.

**✗ Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

**5) D — Implement the corresponding solution: 'replaces telnet'=SSH (remote secure)....**

**✓ Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Secure Protocols): 'replaces telnet'=SSH (remote secure).

**✗ Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

**6) C — Implement the corresponding solution: 'secure remote protocol on port 22' = SSH; not Telnet....**

**✓ Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Secure Protocols): 'secure remote protocol on port 22' = SSH; not Telnet. SSH (Secure

Shell) provides encrypted remote access, while Telnet transmits data in plaintext, making it insecure.

**✗ Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

**7) B — Implement the corresponding solution: Secure ports: SSH 22, HTTPS 443, SFTP, SNMPv3, LDAPS 636....**

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Secure Protocols): Secure ports: SSH 22, HTTPS 443, SFTP, SNMPv3, LDAPS 636. These protocols provide encrypted communication: SSH for secure remote administration, HTTPS for secure web traffic, SFTP for secure file transfer, SNMPv3 for secure network management, and LDAPS for secure directory services.

**✗ Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and it breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials*: This destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

**8) D — Implement the corresponding solution: INSECURE (prohibited on the exam): Telnet 23, FTP 21, HTTP 80, SNMPv1/v2....**

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Secure Protocols): INSECURE (prohibited on the exam): Telnet 23, FTP 21, HTTP 80, SNMPv1/v2.

**✗ Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

**9) C — Implement the corresponding solution: 'replaces telnet'=SSH (remote secure)....**

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Secure Protocols): 'replaces telnet'=SSH (remote secure). SSH provides encrypted remote administration, replacing insecure Telnet which transmits data in cleartext.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

**10) A — Implement the corresponding solution: 'secure remote protocol on port 22'=SSH; not telnet.....**

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Secure Protocols): 'secure remote protocol on port 22'=SSH; not telnet. SSH operates on TCP port 22 and provides encrypted remote access, unlike Telnet which is insecure.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.